Ali Reza Ibrahimzada

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Research Interest

My research covers software engineering, machine learning, artificial intelligence, LLM agents, LLMs for code, and formal reasoning with the goal of modernizing and enhancing the reliability of large-scale software systems. To this end, I leverage program analysis and LLMs to automatically translate and validate repository-level code across different programming languages. So far, my work has focused on evaluating LLMs in code translation and investigating translation bugs [C6], source code decomposition for code translation [C7], and building end-to-end neuro-symbolic [C5] and agentic [C2] repository-level code translation and validation [C1, C3] tools.

Beside code translation, I have also enjoyed working on other problems in software engineering [C4, C8] and machine learning [J1, J2, J3].

Education

08/22 – 08/26 **Ph.D. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Neuro-Symbolic Code Translation and Validation*

08/22 – 12/24 **M.Sc. Computer Science, University of Illinois Urbana-Champaign, IL, USA**
GPA: 4.00 / 4.00
Advisor: Reyhaneh Jabbarvand
Thesis: *Bridging the Gap between Testing and Debugging through Explainable Deep Oracles*

09/18 – 07/22 **B.Sc. Computer Engineering, Marmara University, Istanbul, Turkey**
GPA: 3.98 / 4.00
Advisor: Mehmet Kadir Baran
Thesis: *Depth Estimation of Stereo Images using Deep Learning*

Professional Experience

08/22 – Present **Research Assistant,** University of Illinois Urbana-Champaign, IL, USA
Advised by: Reyhaneh Jabbarvand
Topic: Code LLMs for Source Code Generation, Understanding and Translation
Responsibilities:

- Performing research in the intersection of large language models and software engineering.
- Evaluating large language models in code translation.
- Building neuro-symbolic tools for unified translation and validation of repository-level code.
- Creating agentic architectures for translation validation and repair.
- Developing agentic workflows for end-to-end whole-repository translation and validation.

Professional Experience (continued)

05/25 – 08/25 **Applied Scientist Intern,** Amazon Web Services, VA, USA
Hosted by: Brandon Paulsen and Joey Dodds
Topic: LLM Agents for Autonomous Translation Validation and Repair
Responsibilities:

- Building MatchFixAgent, a language-agnostic and agentic technique for translation validation and repair.
- Creating benchmarks from existing techniques and evaluating MatchFixAgent.
- Performing cost analysis of MatchFixAgent against existing language-specific techniques.

05/24 – 08/24 **Research Intern,** IBM Research, NY, USA
Hosted by: Saurabh Sinha and Raju Pavuluri
Topic: End-to-End Repository-level Code Translation and Validation
Responsibilities:

- Developing AlphaTrans, the first end-to-end neuro-symbolic tool for repository-level code translation and validation.
- Creating a benchmark of real-world Java repositories and evaluating AlphaTrans on them.
- Developing a language interoperability approach for validating translations in isolation.

05/23 – 08/23 **Research Intern,** IBM Research, NY, USA
Hosted by: Saurabh Sinha and Raju Pavuluri
Topic: Evaluating LLMs in Code Translation and Bug Analysis
Responsibilities:

- Performing a large-scale empirical study on LLM-based code translation.
- Performing a large-scale human study on translation bugs generated by LLMs.
- Comparing LLMs with transpilers in code translation.

05/21 – 08/22 **Research Intern,** University of Illinois Urbana-Champaign, IL, USA
Hosted by: Reyhaneh Jabbarvand
Topic: Neural and Interpretable Test Oracles using Deep Learning
Responsibilities:

- Building SEER, a deep neural model for test oracle problem.
- Evaluating SEER on the Defects4J benchmark.
- Comparing SEER with traditional mutation-based techniques.

06/19 – 02/22 **Research Intern,** Istanbul Technical University, Istanbul, Turkey
Hosted by: Ali Cakmak
Topic: Machine Learning
Responsibilities:

- Research on using machine learning techniques for scalable classification of living organisms.
- Developing machine learning techniques for predicting future COVID mutants.
- Developing tools for predicting student success using machine learning.

Research Publications

Conference Proceedings

[C1] K. Ke, **A. R. Ibrahimzada**, R. Pan, S. Sinha, and R. Jabbarvand, "Mockingbird: Context-aware repository-level code translation with isolated validation," in *2027 IEEE/ACM 49th International Conference on Software Engineering (ICSE)*, 2027. URL: https://arxiv.org/abs/2511.21878

[C2] **A. R. Ibrahimzada**, B. Paulsen, D. Kroening, and R. Jabbarvand, "Recodeagent: A multi-agent workflow for language-agnostic translation and validation of large-scale repositories," in *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering*, ser. ASE '26, Munich, Germany: Association for Computing Machinery, 2026, pp. 2409–2421, ISBN: 9798400728822. DOI: 10.1145/3832783.3837486

[C3] **A. R. Ibrahimzada**, B. Paulsen, R. Jabbarvand, J. Dodds, and D. Kroening, "MatchFixAgent: Language-agnostic autonomous repository-level code translation validation and repair," in *Proceedings of the 43rd International Conference on Machine Learning*, T. Zhang, M. Dudik, M. Jaggi, A. Agarwal, S. Li, D. Schuurmans, J. Zhu, F. Berkenkamp, H. Dong, and A. Bietti, Eds., ser. Proceedings of Machine Learning Research, vol. 306, PMLR, Jun. 2026, pp. 49 428–49 451. URL: https://proceedings.mlr.press/v306/ibrahimzada26a.html

[C4] **A. R. Ibrahimzada**, Y. Chen, R. Rong, and R. Jabbarvand, "Challenging bug prediction and repair models with synthetic bugs," in *2025 IEEE International Conference on Source Code Analysis & Manipulation (SCAM)*, 2025, pp. 133–144. DOI: 10.1109/SCAM67354.2025.00021

[C5] **A. R. Ibrahimzada**, K. Ke, M. Pawagi, M. S. Abid, R. Pan, S. Sinha, and R. Jabbarvand, "Alphatrans: A neuro-symbolic compositional approach for repository-level code translation and validation," FSE, vol. 2, New York, NY, USA: Association for Computing Machinery, Jun. 2025. DOI: 10.1145/3729379

[C6] R. Pan*, **A. R. Ibrahimzada***, R. Krishna, D. Sankar, L. P. Wassi, M. Merler, B. Sobolev, R. Pavuluri, S. Sinha, and R. Jabbarvand, "Lost in translation: A study of bugs introduced by large language models while translating code," in *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*, ser. ICSE '24, Lisbon, Portugal: Association for Computing Machinery, 2024, ISBN: 9798400702174. DOI: 10.1145/3597503.3639226

[C7] **A. R. Ibrahimzada**, "Program decomposition and translation with static analysis," in *Proceedings of the 2024 IEEE/ACM 46th International Conference on Software Engineering: Companion Proceedings*, ser. ICSE-Companion '24, Lisbon, Portugal: Association for Computing Machinery, 2024, pp. 453–455, ISBN: 9798400705021. DOI: 10.1145/3639478.3641226

[C8] **A. R. Ibrahimzada**, Y. Varli, D. Tekinoglu, and R. Jabbarvand, "Perfect is the enemy of test oracle," in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ser. ESEC/FSE 2022, Singapore, Singapore: Association for Computing Machinery, 2022, pp. 70–81, ISBN: 9781450394130. DOI: 10.1145/3540250.3549086

Journal Articles

[J1] **A. R. Ibrahimzada**, K. Kosif, A. S. Gulsen, Y. S. Yaglica, and A. Cakmak, "Predicting student grades via adaptive multi-level learning models," *Scientific Reports*, 2026. URL: https://doi.org/10.1038/s41598-026-55750-z

[J2] A. Cakmak, H. Ayaz, S. Arıkan, **A. R. Ibrahimzada**, Ş. Demirkol, D. Sönmez, M. T. Hakan, S. T. Sürmen, C. Horozoğlu, M. B. Doğan, Ö. Küçükhüseyin, C. Cacına, B. Kıran, Ü. Zeybek, M. Baysan, and İ. Yaylım, "Predicting the predisposition to colorectal cancer based on snp profiles of immune phenotypes using supervised learning models," *Medical & Biological Engineering & Computing*, vol. 61, no. 1, pp. 243–258, Jan. 2023, ISSN: 1741-0444. DOI: 10.1007/s11517-022-02707-9

[J3] G. N. Sohsah, **A. R. Ibrahimzada**, H. Ayaz, and A. Cakmak, "Scalable classification of organisms into a taxonomy using hierarchical supervised learners," *Journal of Bioinformatics and Computational Biology*, vol. 18, no. 05, p. 2 050 026, 2020, PMID: 33125294. DOI: 10.1142/S0219720020500262 eprint: https://doi.org/10.1142/S0219720020500262.

Thesis

[T1] **A. R. Ibrahimzada**, *Neuro-symbolic code translation and validation*, 2026.

[T2] **A. R. Ibrahimzada**, *Bridging the gap between testing and debugging through explainable deep oracles*, 2024.

Awards

2024 **Nomination for the Two Sigma PhD Fellowship**, University of Illinois Urbana-Champaign
3rd in Student Research Competition @ ICSE'24, ACM
Travel Grant to attend ICSE'24 in Lisbon, Portugal, NSF
2022 **Travel Grant to attend ESEC/FSE'22 in Singapore, Singapore**, ACM
Valedictorian of Class of 2022, Marmara University
Ray Ozzie Computer Science Fellowship, University of Illinois Urbana-Champaign
Best Senior Graduation Project of the Year, Marmara University
2019 **Academic Achievement Scholarship**, Marmara University
2017 **Valedictorian of High School**, KEN
2015 **Travel Grant to attend QUEST'15 in Lucknow, India**, KEN

Talks

ASE'26 ReCodeAgent: A Multi-agent Workflow for Language-Agnostic Translation and Validation of Large-Scale Repositories, Munich, Germany
ICML'26 MatchFixAgent: Language-Agnostic Autonomous Repository-Level Code Translation Validation and Repair, Seoul, South Korea
SCAM'25 Challenging Bug Prediction and Repair Models with Synthetic Bugs, Auckland, New Zealand
FSE'25 AlphaTrans: A Neuro-Symbolic Compositional Approach for Repository-Level Code Translation and Validation, Trondheim, Norway
ICSE'24 Lost in Translation: A Study of Bugs Introduced by Large Language Models while Translating Code, Lisbon, Portugal
FSE'22 Perfect Is the Enemy of Test Oracle, Singapore, Singapore

Skill Stack

Programming Languages Python, Java, JavaScript, Rust, C, OCaml
Common Docker, Vim, Git, SQL
Machine Learning PyTorch, TensorFlow, vLLM, Codex, Claude Code

Service

Program Committee NeurIPS'26 (Sydney, Australia), ICML'26 (Seoul, South Korea), ICLR'26 (Rio de Janeiro, Brazil), ICLR'25 (Singapore, Singapore), MSR'25 (Ottawa, Canada), MSR'24 (Lisbon, Portugal)
Artifact Evaluation ICSE'25 (Ottawa, Canada)
Organizing ReCode@ICSE'27 (Dublin, Ireland), ReCode@ICSE'26 (Rio de Janeiro, Brazil)

Teaching Experience

Fall 2025 CS 421 - Programming Languages and Compilers w/ Elsa Gunter
Spring 2025 CS 447 - Natural Language Processing w/ Julia Hockenmaier

References

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